

ROCK PAPER SCISSORS

Project Report

A Python-Based Interactive Game

Submitted as a University Programming Project

1. Introduction

Rock Paper Scissors is a simple interactive game in which a player competes against a computer. The player selects Rock, Paper, or Scissors, while the computer generates its choice. The program compares both choices and determines whether the player wins, loses, or draws. This project implements the game in Python using a modular structure and provides additional features such as score tracking and game history.

2. Objectives

The main objectives of this project are to develop an interactive Python game and demonstrate fundamental programming concepts. The project focuses on modular programming, user input, conditional logic, random computer choices, score management, file handling, and basic testing.

3. Problem Statement

The aim is to design a console-based Rock Paper Scissors game that allows a user to play against a computer, determines the result correctly, maintains the scoreboard, and stores previous game results for later viewing.

4. Features

The project provides the following features:

- Play Rock Paper Scissors against the computer.
- Generate a computer choice.
- Determine win, loss, or draw.
- Maintain and display the scoreboard.
- Save and display previous game history.
- Display the rules through the menu.
- Provide an option to exit the program.

5. Technologies Used

The project is implemented using Python. It uses Python modules for separating different responsibilities of the application. File handling is used for maintaining game history, and unit testing is used to test the game logic.

6. Project Structure

The project is organized into multiple files. The main program connects the different modules and provides the interactive menu. Separate modules handle the computer choice, game result, player input, scoreboard, history, and testing.

File	Purpose
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main.py	Main program and interactive menu
game.py	Game result and winning logic
computer.py	Computer choice generation
utils.py	Player input and utility functions
scoreboard.py	Score and statistics management
history.py	Saving and displaying game history
test_game.py	Testing the game logic

7. Working and Methodology

When the program starts, the user is shown an interactive menu. The user can start a game, view the scoreboard, view game history, read the rules, or exit. During a game, the player's choice and the computer's choice are obtained. The game logic compares the two choices and produces the result. The scoreboard is then updated and the game details are saved to the history.

8. Game Rules

The rules implemented in the project are straightforward: Rock beats Scissors, Scissors beats Paper, Paper beats Rock, and identical choices result in a draw.

9. Testing

The project includes a separate test file for checking the game logic. Testing helps verify that different combinations of Rock, Paper, and Scissors produce the expected results. The test module can be executed using Python's unittest framework.

10. Advantages

The project has a simple user interface, clear modular organization, reusable components, score tracking, and persistent game history. Its structure also makes individual parts easier to understand and test.

11. Limitations

The current project is console-based and does not provide a graphical interface. The game is designed for basic single-player interaction against the computer.

12. Future Scope

The project can be extended by adding a graphical user interface, multiple rounds, difficulty levels, additional statistics, player profiles, improved data storage, and a more advanced user interface.

13. Conclusion

The Rock Paper Scissors project demonstrates how Python programming concepts can be combined to create a complete interactive application. Through modular design, game logic, score management, history storage, and testing, the project provides practical experience with fundamental software development concepts.

14. References

Python documentation and standard Python programming concepts were used as references while developing the project.